

Don't Just Listen, Try Planning

Graph-based Retrieval-Generation Agent for Long-form Audio Meeting Understanding

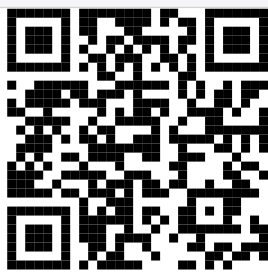
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[LongAudioQA Dataset](#) [Multi-dimensional Graph](#) [Agent Planning](#) [Acoustic-aware QA](#)

Code & Data

Scan for GRGA resources and
LongAudioQA.

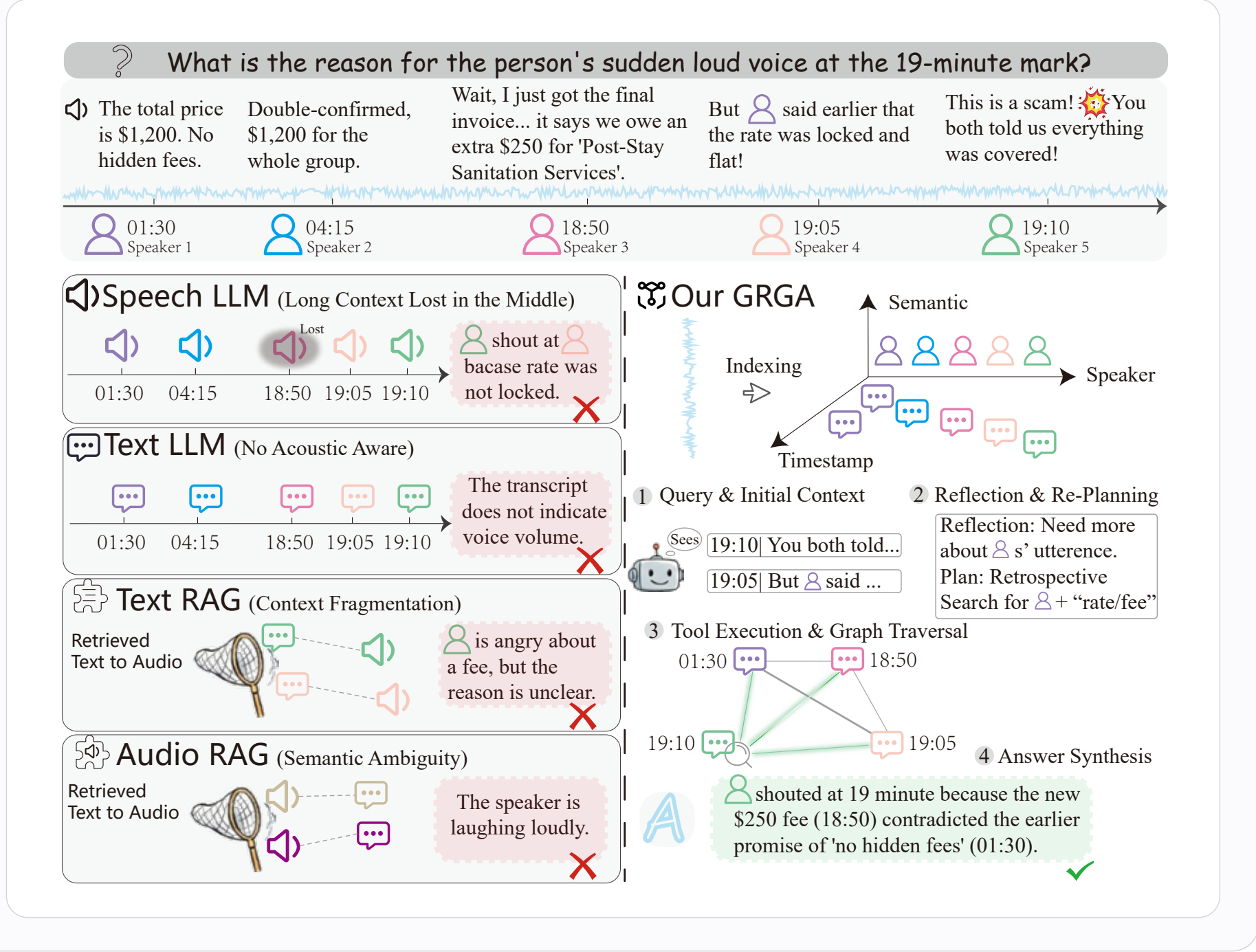


Takeaway:

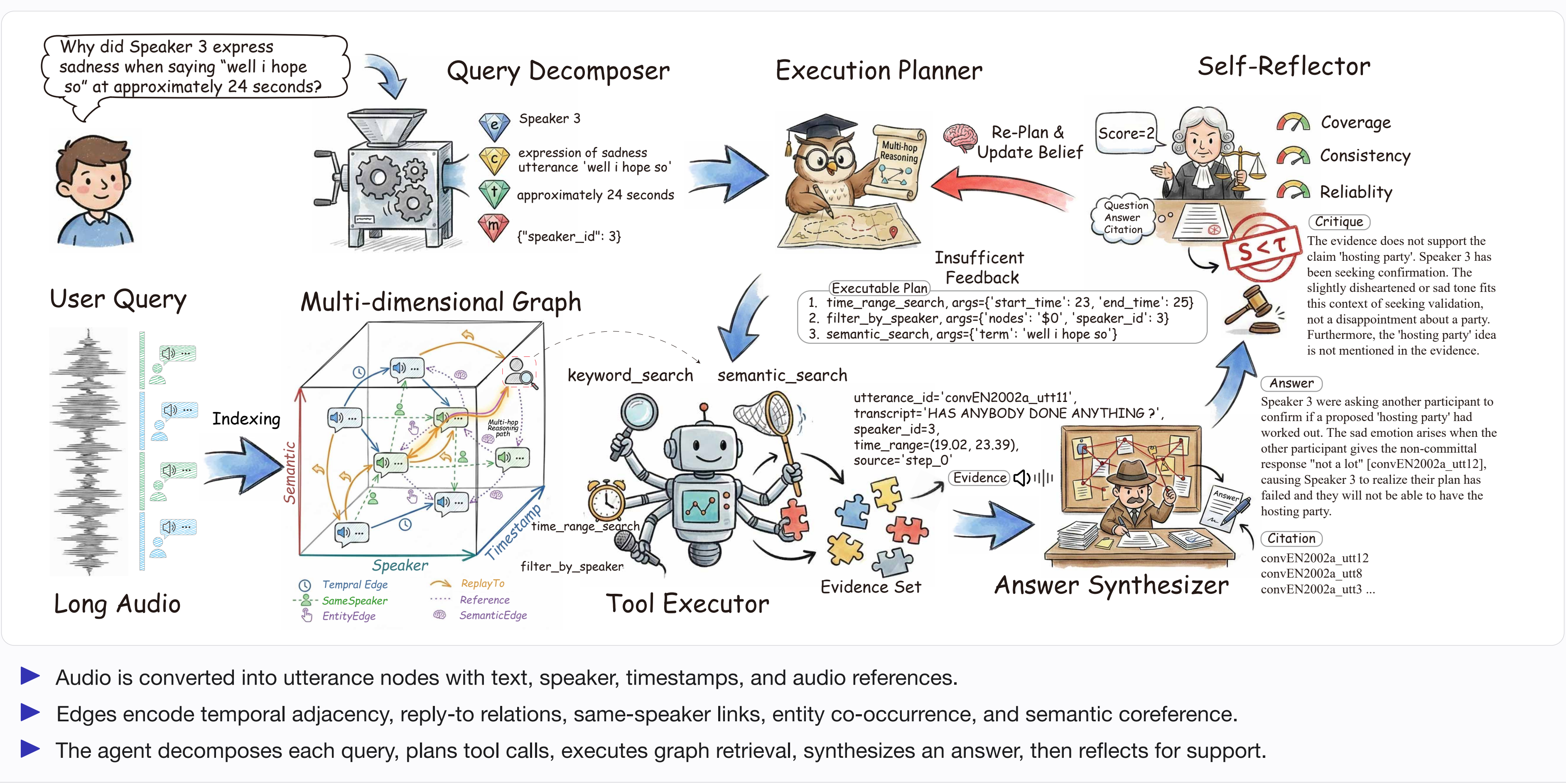
Plan over semantic, speaker, timestamp, and acoustic evidence instead of retrieving one flat chunk.

Why Long Audio QA Breaks

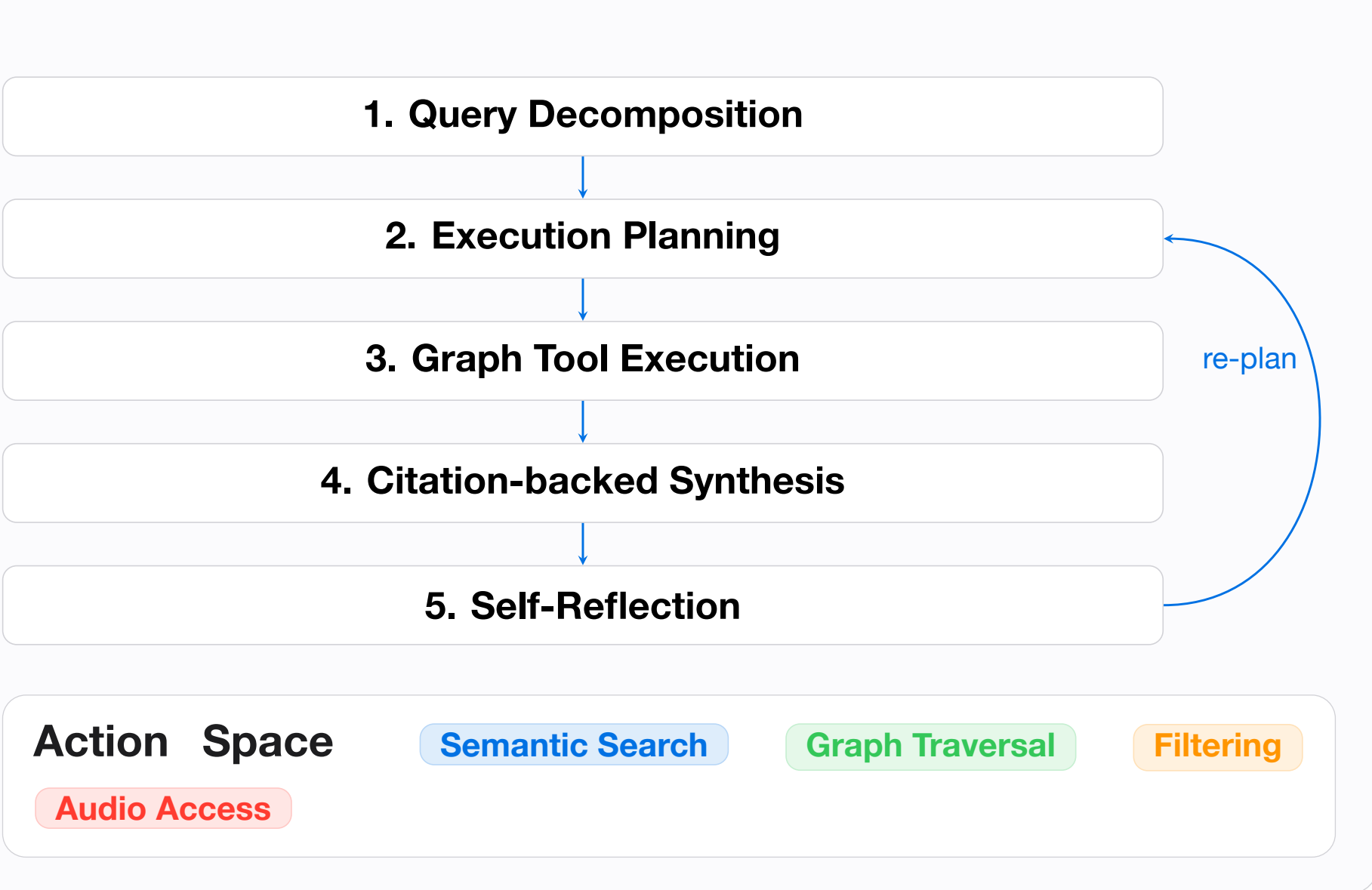
- Long-form audio meeting understanding lacks dedicated QA benchmarks.
- Speech LLMs map audio toward text, losing paralinguistic cues such as loudness, emotion, or speaker tension.
- Flat RAG retrieves local transcript chunks, but meeting answers often require long-range evidence and speaker/time links.



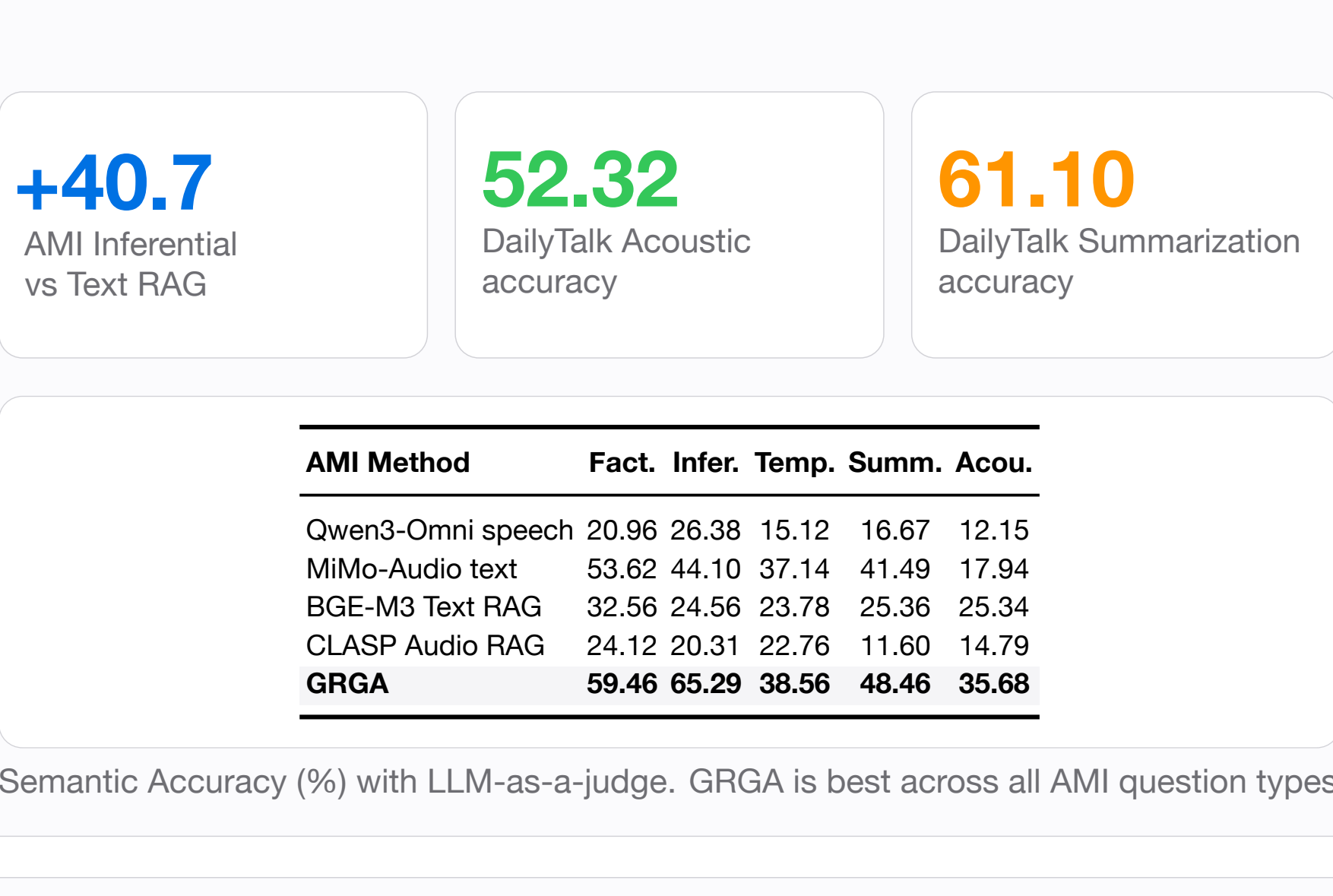
GRGA: Graph Retrieval + Planning



Reasoning Loop

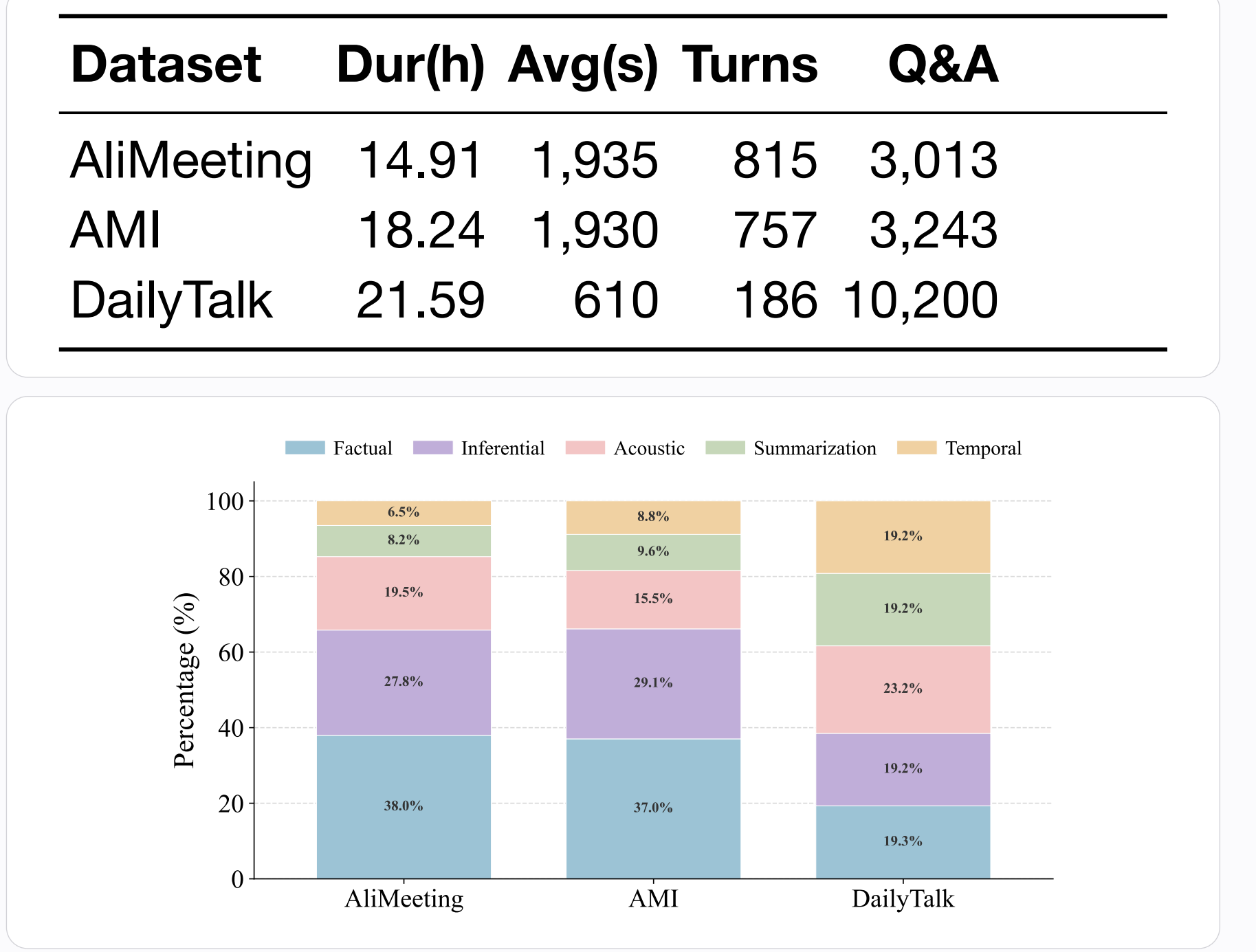


Main Results



LongAudioQA

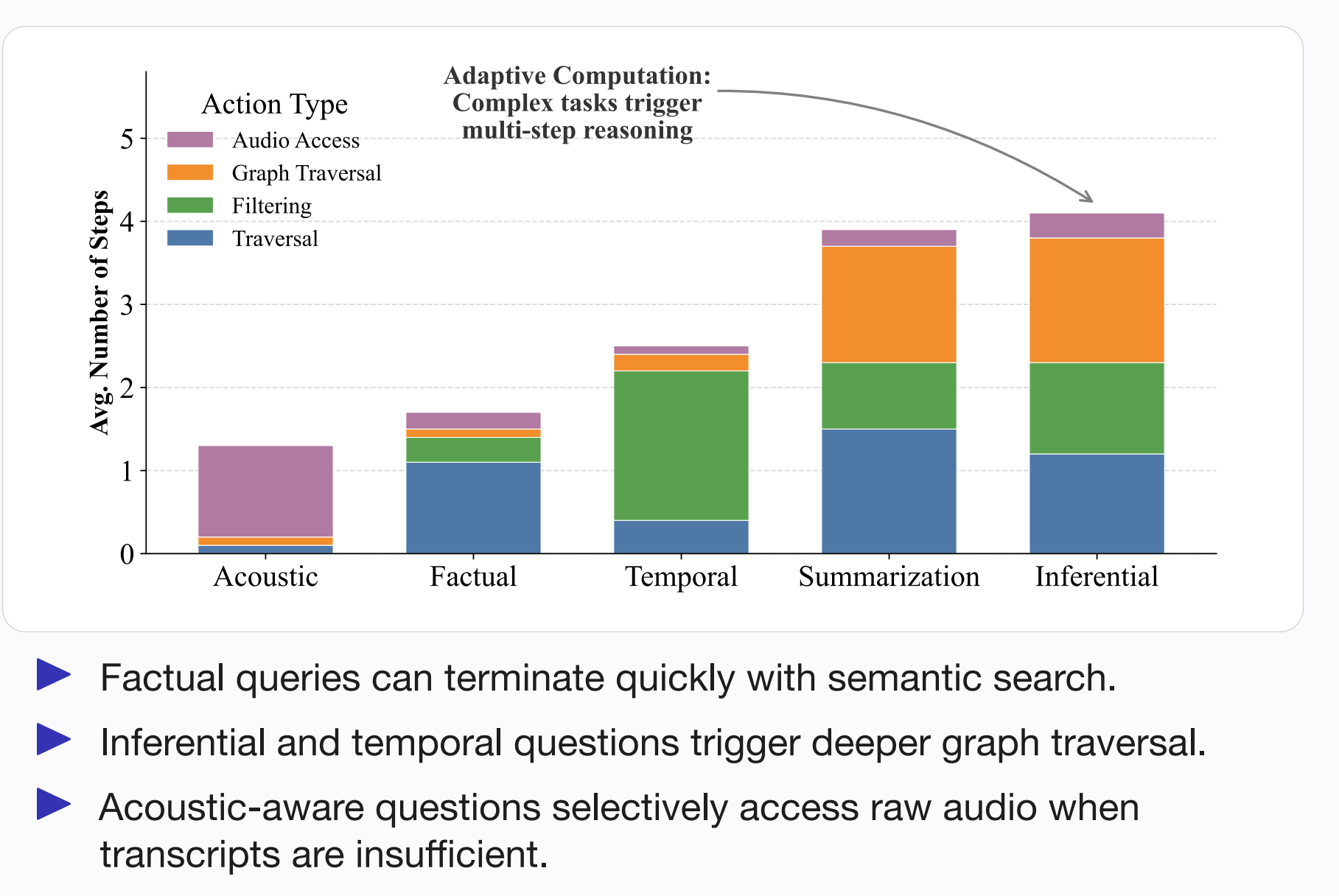
- New QA dataset for long-form meeting and conversation understanding.
- Built from AliMeeting, AMI Meeting, and DailyTalk with human verification.
- Five skills: factual retrieval, inferential reasoning, temporal grounding, summarization, and acoustic-aware reasoning.



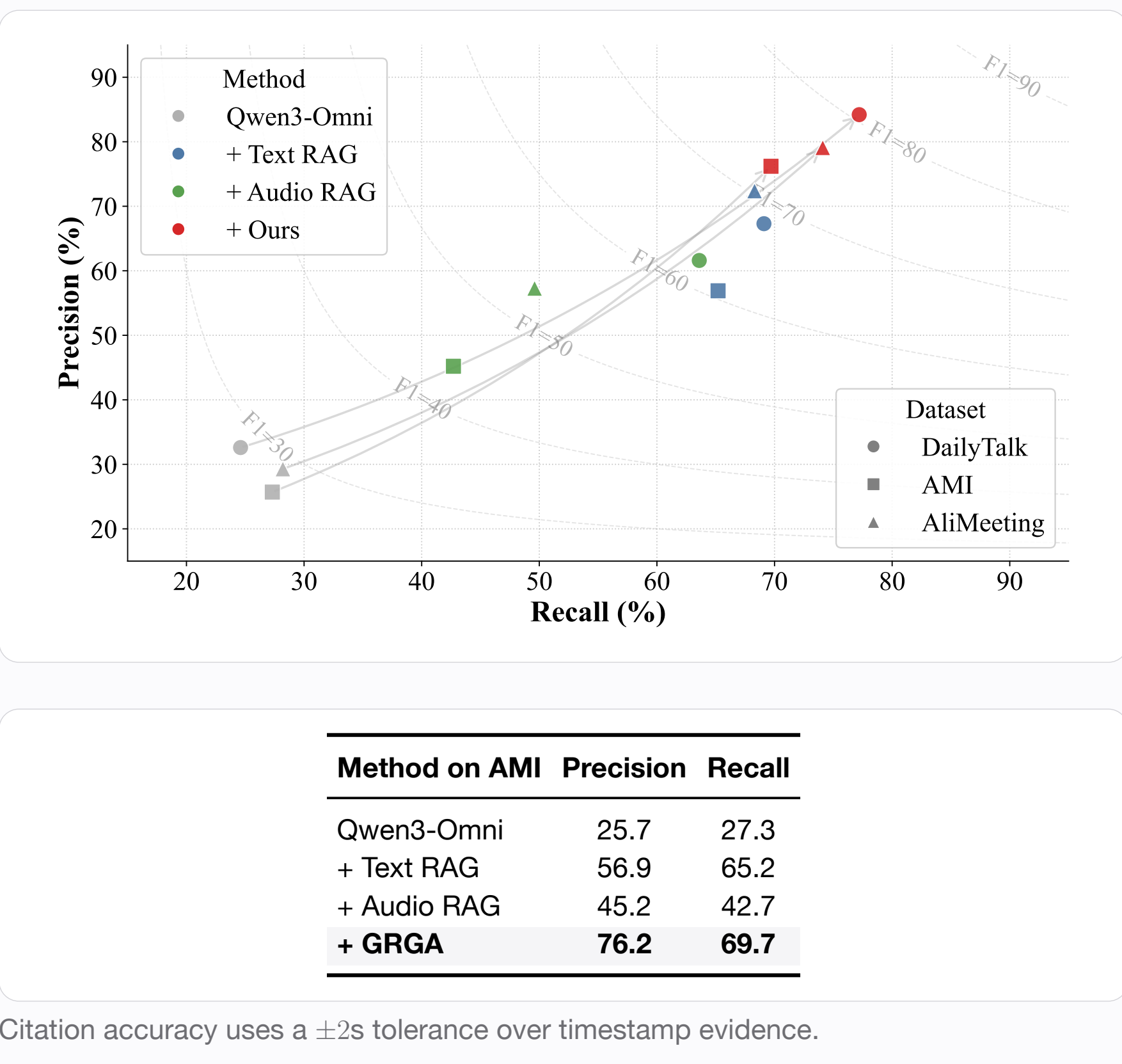
Question Taxonomy

Factual	Entity retrieval, attribute association, keyword locating.
Inferential	Multi-hop links between early problems, later discussions, and final decisions.
Temporal	Absolute localization, relative sequencing, frequency statistics in time windows.
Summarization	Topic consensus and speaker contribution aggregation.
Acoustic-aware	Align emotion, intensity, laughter, or volume spikes with semantic context.

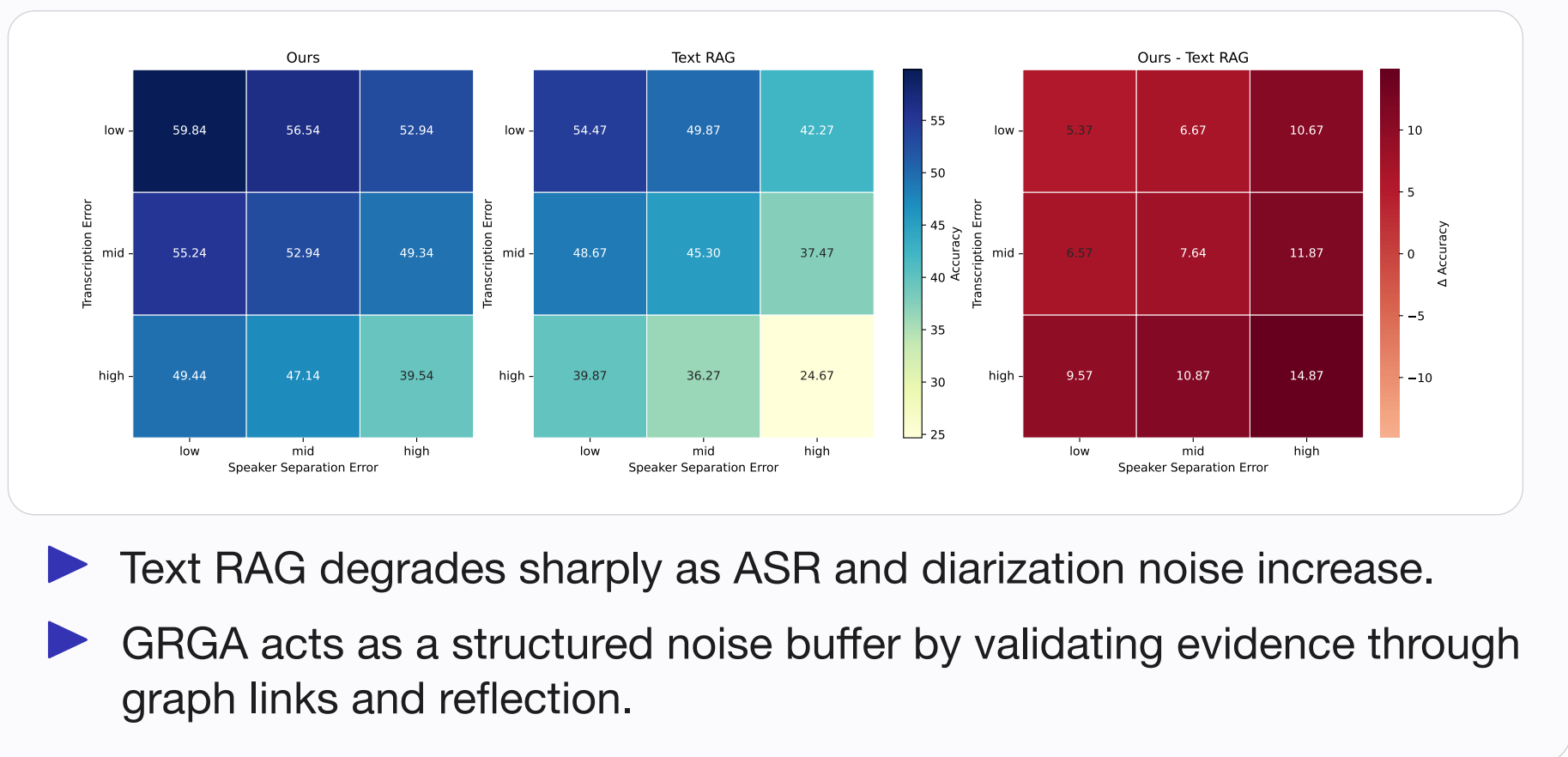
Why Planning Helps



Grounded Answers



Robustness Under Noise



Conclusion

Long-form audio QA needs planning, not just listening.

- Dataset:** LongAudioQA introduces long-context, multi-speaker, temporal, and acoustic-aware QA.
- Method:** GRGA turns meetings into multi-dimensional graphs and solves QA through tool-using agent planning.
- Evidence:** Improvements appear in accuracy, citation precision/recall, human groundedness, and noisy settings.

Self-Correction Case

Query: "What is the team's general attitude to the project deadline and workload?"	
Initial answer	Generic summary: the deadline is tight and members feel urgency.
Reflection	Low confidence; needs concrete evidence for "tight" and "work-intensive".
Re-plan	Search for deadline and workload sentiment; retrieve specific utterances.
Final answer	Cites "six days", "change the deadlines", "a hassle", and "superficially".

Reflection converts unsupported fluent text into evidence-seeking behavior.

Final response is anchored to exact utterance-level citations.

Ablation Highlights

Removed Component	Avg.	Drop
Full GRGA	49.49	-
w/o Reflection	39.64	-9.84
w/o Graph Traversal	38.21	-11.27
w/o Audio Access	34.71	-14.78
w/o Semantic Search	16.28	-33.21

Semantic search finds entry evidence; graph traversal connects dispersed clues.

Audio access is crucial for acoustic-aware questions.

Reflection prevents accepting unsupported first-pass answers.

Human Evaluation

